

XI LI

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Assistant Professor, Department of Computer Science, University of Alabama at Birmingham
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RESEARCH INTERESTS

Trustworthy AI – Addressing ethical challenges in AI systems through interdisciplinary frameworks. My work aims to develop reliable AI technologies that align technical innovation with societal well-being.

EMPLOYMENT

Tenure-Track Assistant Professor Department of Computer Science, UAB	Aug. 2024 – present
Associate Scientist Integrative Center for Aging Research, School of Medicine, UAB	Sep. 2025 – present
Associate Scientist Center for the Study of Community Health, School of Public Health, UAB	Dec. 2025 – present
Machine Learning Engineer Intern Modern Recommendation System Team, Meta	May 2024 – Aug. 2024

EDUCATION

The Pennsylvania State University Ph.D. in Computer Science and Engineering Advisor: Dr. George Kesidis and Dr. David Miller M.S. in Computer Science and Engineering	State College, PA Aug. 2018 - Aug. 2024 Aug. 2016 - July 2018
The Southeast University B.E. in Electrical Engineering	Nanjing, China Aug. 2012 - June 2016

PUBLICATIONS

- **X. Li**, R. Mao, Y. Zhang, R. Lou, C. Wu, J. Wang. *Chain-of-Scrutiny: Detecting Backdoor Attacks for Large Language Models*. ACL findings 2025.
- **X. Li**, S. Zhao, F. Zhao, R. Yu. *Rethinking the Safety Landscape for Foundation Models: A Multi-Modal Perspective*. ICCV T2FM, 2025.
- **X. Li**, J. Wang. *Federated Foundation Models Raise New Concerns of Robustness, Privacy, and Fairness*. ICCV T2FM, 2025.
- **X. Li**, C. Wu, J. Wang. *Vulnerabilities of Foundation Model Integrated Federated Learning Systems Under Adversarial Threats*. IJCNN, 2025.
- F. Fan, **X. Li**. *PeerGuard: Defending Multi-Agent Systems Against Backdoor Attacks Through Mutual Reasoning*. IEEE IRI Best Paper Award, 2025.
- X. Bi, **X. Li**. *Securing Federated Learning Against Novel and Classic Backdoor Threats During Foundation Model Integration*. IEEE IRI, 2025.
- G. Yang, **X. Li**, H. Wang, D. J. Miller, G. Kesidis. *Backdoor Inversion in Neural-Activation Space*. MLSP, 2025.
- F. Zhao, C. Zhang, R. Zhang, T. Wang, **X. Li**, *Mitigating Image Captioning Hallucinations in Vision-Language Models*. IEEE MIPR, 2025.

- **X. Li**, Z. Xiang, D. J. Miller, G. Kesidis. *Correcting the distribution of batch normalization signals for Trojan mitigation*. Neurocomputing, 2024.
- **X. Li**, D. J. Miller, Z. Xiang, G. Kesidis. *BIC-based Mixture Model Defense against Data Poisoning Attacks on Classifiers: A Comprehensive Study*. IEEE Transactions on Knowledge and Data Engineering (TKDE), 2024.
- **X. Li**, C. Wu, J. Wang. *Unveiling Backdoor Risks Brought by Foundation Models in Heterogeneous Federated Learning*. The Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD), 2024.
- **X. Li**, S. Wang, R. Huang, M. Gowda, G. Kesidis. *Temporal-Distributed Backdoor Attack Against Video Based Action Recognition*. AAAI, 2024.
- **X. Li**, S. Wang, C. Wu, H. Zhou, J. Wang. *Backdoor Threats from Compromised Foundation Models to Federated Learning*. FL@FM-NeurIPS'23.
- **X. Li**, D. J. Miller, Z. Xiang, G. Kesidis. *A BIC-based Mixture Model Defense against Data Poisoning Attacks on Classifiers*. IEEE International Workshop on Machine Learning for Signal Processing (MLSP), 2023.
- **X. Li**, D. J. Miller, Z. Xiang, G. Kesidis. *Test-Time Detection of Backdoor Triggers of Poisoned Deep Neural Networks*. IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2022.
- Z. Xiang, D. J. Miller, S. Chen, **X. Li**, G. Kesidis. *Detecting Backdoor Attacks Against Point Cloud Classifiers*. IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2022.
- Z. Xiang, D. J. Miller, S. Chen, **X. Li**, G. Kesidis. *A Backdoor Attack against 3D Point Cloud Classifiers*. ICCV, 2021.

PREPRINTS

- J. Wang, Y. Zhang, **X. Li**. *NeuroGen: Neural Network Parameter Generation via Large Language Models*. under review, 2025.
- Jiaqi Wang, Yusen Zhang, **Xi Li**, Simeng Han, Zining Zhu, Lingjuan Lyu, Yaqing Wang, Ruiquan Huang, Arman Cohan, Rui Zhang, Sheng Li, Xin (Eric) Wang. *Survey on Neural Network Parameter Acquisition: from Optimization to Generation*. 2025

HONORS & SCHOLARLY ACTIVITIES

- **Tutorial Organizer**, *Towards Safe Multi-Modal Learning: Unique Challenges and Future Directions*, ICCV 2025
- **Tutorial Organizer**, *Exploitation and Mitigation: Understanding Large-Scale Machine Learning Robustness under Paradigm Shift*, SDM 2025
- **Best Paper Award**, IEEE IRI, 2025

TEACHING EXPERIENCE

- **Instructor** UAB
 - CS652/752 Advanced Algorithms and Applications 2025, 2026 Spring
 - CS685/785 Foundation of Data Science 2024, 2025 Fall
 - CS300/500 Advanced OO Programming in C++ 2025 Fall
- **Teaching Assistant** PSU
 - CMPSC465 Data Structures and Algorithms 2021 Spring
 - CMPSC/MATH 451 Numerical Computations 2019 Fall - 2024 Spring
 - CMPEN/EE 455 Digital Image Processing I 2018 Fall

PROFESSIONAL SERVICE

- **Proposal Panelist:** NSF
- **Program Committee Member:** AAAI, SDM
- **Conference Reviewer:** AAAI, ICCV, CVPR, NeurIPS, ICLR, AISTATS, KDD, COLM, ICASSP
- **Journal Reviewer:** IEEE Transactions on Neural Networks and Learning Systems (TNNLS), Neurocomputing, Neural Networks, Computers & Security, IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
- **Student Volunteer:** NeurIPS